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Gradient-free learning of a closed-loop wall controller for turbulent drag reduction

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Closed-loop wall control learnt by multi-agent reinforcement learning can lower skin-friction drag in turbulent channels, but these gradient-based policies are trained on small periodic boxes and exhibit reduced performance when carried over to a larger domain. We recently showed that such policies are also prone to saturated bang-bang actuations that collapse into standing streamwise waves whose scale is set by the computational box rather than by the near-wall cycle, and proposed architectural fixes that avoid these degeneracies (Cavallazzi *et al.* 2026). Here, we employ Evolution Strategy (ES) to optimise a recurrent closed-loop controller directly on a large turbulent channel at $Re_\tau \approx 180$, evaluating policy performance over full flow episodes using an energy-aware criterion and processing candidate policies in parallel. To our knowledge, this is the first application of an evolution strategy to the control of a turbulent flow. The ES controller reduces the skin friction by about 26%, exceeding the gradient-based multi-agent controller of Cavallazzi *et al.* (2026) (GRU-MARL), trained on a minimal box (17%), and marginally exceeding classic opposition control (OC, 22%). A wall-normal decomposition of the friction, Reynolds-stress profiles and anisotropy invariants show that the ES and opposition-controlled flows follow separate trajectories through the buffer layer, reaching comparable drag reduction by different reorganisations of the near-wall turbulence. In particular, the ES actuation correlates predominantly with the streamwise velocity fluctuations rather than with the wall-normal velocity that classical OC targets.

1. Introduction

Skin-friction drag in wall-bounded turbulence is generated by quasi-streamwise vortices whose induced wall-normal velocities sustain the Reynolds shear stress above its laminar value (Orlandi & Jiménez 1994). The vortices are part of a self-sustaining near-wall cycle of low-speed streaks and quasi-streamwise vortices (Kline *et al.* 1967; Jiménez & Moin 1991; Hamilton *et al.* 1995; Waleffe 1997) that persists autonomously within the buffer layer even in the absence of outer-flow forcing (Jiménez & Pinelli 1999), making that layer the natural target for drag-reduction control.

In the case of established active flow control methods, like opposition control (Choi *et al.* 1994; Hammond *et al.* 1998; Stroh *et al.* 2015), spanwise forcing and streamwise-travelling waves of spanwise velocity (Quadrio *et al.* 2009; Gatti & Quadrio 2016; Toubert & Leschziner 2012), the drag-reduction mechanism rests on a specific, physically motivated interaction with the near-wall structures, and the actuation is designed around that interaction from the outset. Controllers discovered by learnt data-driven algorithms carry no such prior; their

actuation strategy emerges from the optimisation, and what physical interaction it exploits is not known in advance.

Reinforcement learning (RL) has been the most productive research area for those controllers. Applied to the classical plane channel flow at low Reynolds numbers, it has produced closed-loop wall policies that report drag reductions competitive with the physics-based benchmarks (Guastoni *et al.* 2023; Sonoda *et al.* 2023; Lee *et al.* 2023; Han & Huang 2020; Zhou *et al.* 2025; Varela *et al.* 2022; Cavallazzi *et al.* 2025), as surveyed in several reviews (Garnier *et al.* 2021; Brunton *et al.* 2020; Vignon *et al.* 2023*b*). A common way to keep the formulation tractable is a shared, translationally invariant multi-agent policy (Vignon *et al.* 2023*a*; Guastoni *et al.* 2023), trained with a centralised critic on a small periodic domain, executed independently at each wall patch, then applied to a larger channel. This template underlies numerous subsequent studies (Sonoda *et al.* 2023; Cavallazzi *et al.* 2025; Wälchli *et al.* 2024; Bae & Koumoutsakos 2022; Suárez *et al.* 2024; Rabault & Kuhnle 2019). The drag reduction headline figures reported on minimal domains must, however, be read with care: a policy evaluated in a minimal flow unit can drive the flow towards relaminarisation, where the pressure gradient falls to its laminar value and the reported drag reduction saturates at a value that reflects the box rather than the controller (Cavallazzi *et al.* 2026). A representative comparison therefore requires a domain large enough to keep the flow fully turbulent throughout, the setting adopted here.

Despite the need for a large evaluation domain, training is in practice confined to a small one by the structure of the multi-agent update rather than by simulation cost alone. A gradient step adjusts the shared policy parameters from a per-patch return whose credit is assigned by a central critic, and that critic must resolve each patch within a joint action space that widens as patches are added, so the estimate deteriorates on larger domains. Two inherent features of the wall-control problem tie this restriction down, one corrupting the credit assignment and the other raising its cost. First, because the actuation must inject no net mass, the joint action is projected onto its zero-mean subspace before reaching the wall. Under this projection, the action applied at any single patch becomes dependent on all others, breaking the separability of per-patch credit (Cavallazzi *et al.* 2026). Second, every training episode requires advancing a costly direct numerical simulation, so the wall-clock cost of collecting experience grows with the domain. Consequently, training is conventionally restricted to a minimal flow unit, and the policy is transferred only afterwards.

An evolution strategy removes both constraints. It treats the expected return as a function of the policy parameters. At each generation, a population of perturbed parameter vectors is evaluated over complete flow episodes. Each episode returns a scalar reward. The parameter update is a weighted average of the perturbations, with no backpropagation through the flow and no central critic (Rechenberg 1973; Hansen & Ostermeier 2001; Wierstra *et al.* 2014; Salimans *et al.* 2017). Because whole episodes are scored, the per-agent credit assignment that the zero-net-mass constraint corrupts does not arise. The population evaluations are independent and parallelise across rollouts. The search therefore runs directly on the large evaluation domain, with no small-box surrogate. To our knowledge, this is the first application of an evolution strategy to the control of a turbulent flow.

The evolution-strategy controller provides a setting to address a question that has not been settled on a large domain: whether a learnt policy that reaches opposition-level drag reduction does so through the opposition mechanism or through a different reorganisation of the near-wall cycle. Its actuation is smooth and free of the bang-bang saturation that characterises degenerate controllers (Cavallazzi *et al.* 2026). The controlled flow is therefore interpretable in physical terms. We characterise it with a wall-normal decomposition of the skin friction (Fukagata *et al.* 2002), Reynolds-stress profiles scaled by each controller’s own

friction velocity, the anisotropy invariants of the stress tensor (Lumley & Newman 1977), and the premultiplied near-wall spectra (del Álamo & Jiménez 2003; del Álamo *et al.* 2004).

2. Problem set-up and methodology

2.1. Flow configuration and numerics

We perform direct numerical simulations of an incompressible flow in a plane channel at a constant flow rate by using CaNS (Costa 2018; Costa *et al.* 2021). The solver integrates the unsteady, incompressible Navier-Stokes equations using a low-storage third-order Runge–Kutta advancement within a pressure-projection algorithm. The friction Reynolds number is $Re_\tau = H^+ \simeq 180$. The domain measures $(L_x^+, L_y^+, L_z^+) \simeq (1922, 576, 360)$ in viscous units, equivalently $(L_x/H, L_y/H, L_z/H) = (10.68, 3.2, 2)$, with H the channel half-height. It is discretised on a $256 \times 256 \times 100$ grid with wall-normal points clustered towards the walls. Throughout, u , v and w denote the streamwise, spanwise and wall-normal velocity components, and p the pressure. Control is applied as wall-normal blowing and suction $w_w(x, y, t)$ at the wall, Γ_w , with zero net mass flux enforced at every instant, $\int_{\Gamma_w} w_w \, dx \, dy = 0$. The flow is sensed at a detection plane at $z_d^+ \simeq 14$. The actuation interval is $\Delta t^+ \simeq 5$, longer than the fastest near-wall fluctuations but short compared to the typical streak lifetime.

2.2. Closed-loop controller and the evolution strategy

The controller is a recurrent neural network that maps the velocity field at the detection plane to a wall-normal velocity field at the wall. At each actuation step the hidden state h_t is updated and a new actuation field a_t is produced. The zero-net-mass constraint is enforced as the final layer: the projection $P : a \mapsto a - \bar{a}$ subtracts the spatial mean. The resulting $w_w(\cdot, t) = Pa_t$ therefore satisfies $\int_{\Gamma_w} w_w \, dx \, dy = 0$ at every instant for any parameter vector $\theta \in \mathbb{R}^d$. The same architecture is used by the GRU-MARL controller of Cavallazzi *et al.* (2026), trained by multi-agent gradient descent (Lowe *et al.* 2017) on a minimal flow unit.

Here, the shared parameters θ are optimised directly on the large evaluation domain by an evolution strategy (Rechenberg 1973; Wierstra *et al.* 2014; Salimans *et al.* 2017). The optimisation target is the net energy saving over a complete episode. Let $P_p = |\langle \partial_x p \rangle| U_b$ be the mean pumping power and $W_w = (L_x L_y)^{-1} \int_{\Gamma_w} w_w p \, dx \, dy$ the net power delivered by the controller at the wall. The total dissipation is $\varepsilon = P_p + W_w/H$, and the fitness is $F(\theta) = (\varepsilon_0 - \varepsilon(\theta))/\varepsilon_0$, where ε_0 is the uncontrolled value. Two metrics commonly used to score wall controllers omit the wall power W_w entirely: the bare drag reduction $1 - |\langle \partial_x p \rangle|/|\langle \partial_x p \rangle_0|$, which tracks only the mean pressure gradient, and the cubic surrogate $\sim |w_w|^3$, which estimates actuation cost from the actuation amplitude rather than from the power exchanged with the wall. Because neither metric can register the energy a controller adds through the wall, a controller that injects energy can score well on either while lowering $F(\theta)$ (Fukagata *et al.* 2009; Marusic *et al.* 2021; Cavallazzi *et al.* 2026).

Unlike gradient-based training, the evolution strategy requires no derivative of F with respect to θ . Obtaining one would mean differentiating through both the DNS solver and the recurrent network; instead, each candidate parameter vector is scored by running one complete flow episode and reading off the resulting F , with the simulation and the network treated as a black box. At each generation, n pairs of parameter vectors are drawn by adding mirrored Gaussian perturbations to the current estimate $m \in \mathbb{R}^d$, $\{m \pm \sigma \xi_k\}_{k=1}^n$ with $\xi_k \sim \mathcal{N}(0, \mathbf{I})$, giving $2n$ members evaluated in parallel, each as one DNS episode initialised from the same flow state so that fitness differences reflect θ rather than the initial condition

(Salimans *et al.* 2017). The mean is then advanced along

$$\hat{g} = \frac{1}{2n\sigma} \sum_{k=1}^n (v_k^+ - v_k^-) \xi_k, \quad (2.1)$$

where $v_k^\pm$ is the centred rank of $F(m \pm \sigma \xi_k)$ within the generation: each member's position among the $2n$ fitness values, shifted so the ranks average to zero. This makes the update depend only on the relative ordering of the population and not on the numerical scale of F (Salimans *et al.* 2017). The parameter mean is then updated from $\hat{g}$ using Adam (Kingma & Ba 2014), a standard adaptive-step-size rule. The $2n$ DNS episodes within a generation are mutually independent and can be run concurrently, so given enough parallel resources the wall-clock cost of a generation reduces to that of a single episode.

2.3. Control cases and diagnostics

Four flows are compared: the uncontrolled channel, opposition control, and two learnt controllers that share the architecture of §2.2 and differ only in how their parameters were obtained. Opposition sets the wall velocity to the negative of the wall-normal velocity sensed at the detection plane, $w_w = -w'(z_d)$ (Choi *et al.* 1994). The first learnt controller is trained by gradient-based multi-agent learning on a minimal flow unit, within a centralised-training, decentralised-execution framework (Lowe *et al.* 2017; Foerster *et al.* 2018) constrained to keep the actuation physically coherent, and applied to the evaluation domain without further training (Cavallazzi *et al.* 2026); it is labelled GRU-MARL. The second is the same policy whose parameters are then optimised directly on the evaluation domain by the evolution strategy of §2.2, and is labelled ES. We omit the saturated, memoryless policies whose behaviour motivates a separate study of reward specification (Cavallazzi *et al.* 2026), since the concern here is the mechanism of controllers that reduce drag at low actuation power.

Each controlled flow is characterised by its own friction velocity $u_\tau = \sqrt{\tau_w/\rho}$ (being τ_w the wall shear stress and ρ the density), evaluated after the control has acted, and all wall-scaled quantities below are normalised by this flow-specific value.

The contribution of the Reynolds shear stress to the skin friction is quantified with the identity of Fukagata *et al.* (2002), which decomposes the friction coefficient $C_f = \tau_w/(\frac{1}{2}\rho U_b^2)$, with $U_b = 1$ constant flow rate, into a laminar contribution C_f^{lam} at the same bulk Reynolds number and a wall-normal weighted integral of the Reynolds shear stress,

$$C_f = C_f^{\text{lam}} + c \int_0^1 (1 - z/H) (-\langle u'w' \rangle) d(z/H), \quad (2.2)$$

where the integral runs from the wall ($z/H = 0$) to the centreline ($z/H = 1$) and the numerical prefactor of the plane-channel identity is $c = 12$ (Fukagata *et al.* 2002). Scale information is taken from the premultiplied spectra of the velocity fluctuations at fixed wall-normal distance (del Álamo *et al.* 2004).

3. Results

The comparison begins with the drag reduction itself. Optimised directly on the evaluation domain, the evolution-strategy controller reduces the skin friction by 26.1%, above the 17.4% obtained by the same architecture trained by gradient-based multi-agent learning on the minimal box and transferred without further training, and marginally above the 22.5% of opposition control. The gap between the two learnt controllers is the drag reduction lost when the gradient-based policy is transferred from its minimal training box to the larger

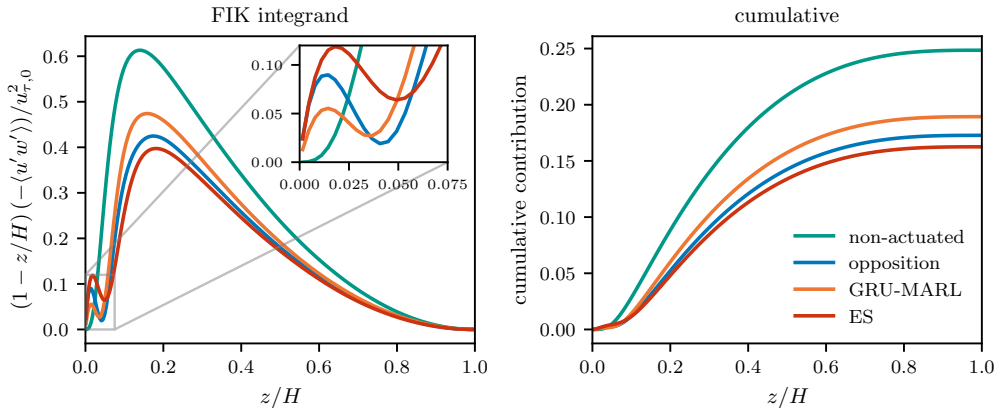


Figure 1: FIK decomposition of the skin friction: the weighted, normalised Reynolds stress (left) and its cumulative integral (right) against the wall-normal coordinate z/H .

channel, and recovered when the evolution strategy optimises on that channel directly. The friction velocities differ between the cases, falling from $u_{\tau} \approx 0.064$ in the uncontrolled flow to 0.056 for opposition, 0.058 for GRU-MARL and 0.055 for the evolution strategy. Reaching comparable drag reduction is the starting point of the comparison; the rest of the paper asks whether the underlying flow states coincide.

Figure 1 decomposes the skin friction across the channel with the identity (2.2), plotted against the wall-normal coordinate z/H and scaled for every case by the single friction velocity of the uncontrolled flow, so that the area between the uncontrolled curve and a controlled one is that controller’s drag reduction. The controlled flows carry a smaller weighted Reynolds stress than the uncontrolled flow across the buffer layer and the region above it, and the cumulative curves show the reduction accruing over the same range of z/H for all three. The relative split between layers is similar across the controllers, the buffer layer carrying about a quarter of the weighted stress in each case, while the turbulent fraction of the mean-energy budget falls from 0.41 in the uncontrolled flow to about 0.34 to 0.35 under control. The decomposition locates the reduction but does not on its own distinguish the controllers; the difference between them lies in how the near-wall stress is organised, which the remaining diagnostics address.

3.1. Reynolds-stress reorganisation

Figure 2 shows the streamwise normal stress $\langle u'u' \rangle^+$ and the Reynolds shear stress $\langle u'w' \rangle^+$, each normalised by its own friction velocity. The shear stress, which carries the drag, follows opposition for both learnt controllers, yet their streamwise normal stress differs from that of every other case, opposition included. The learnt controllers therefore reach an opposition-like shear stress through a streamwise fluctuation field that no single benchmark reproduces.

Figure 3 maps the same information onto the anisotropy-invariant triangle. The trajectories of the controlled flows through the buffer layer occupy different regions of the triangle: a coordinate-free confirmation that the controlled states are distinct and that depart differently from the uncontrolled trajectory. This is consistent with the stress profiles of figure 2.

3.2. Near-wall structure, actuation and scale selectivity

Figure 4 shows instantaneous fields of the near-wall velocity and of the wall actuation for each controller. Opposition sets the wall velocity to the negative of the sensed wall-normal

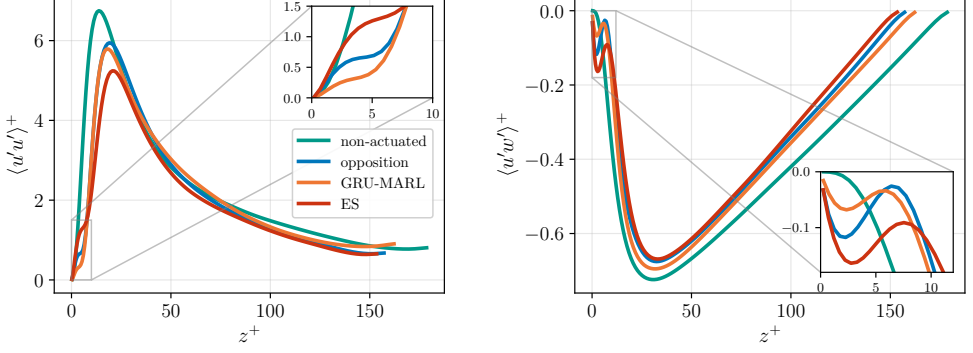


Figure 2: Streamwise normal stress $\langle u'u' \rangle^+$ (left) and Reynolds shear stress $\langle u'w' \rangle^+$ (right), each renormalised by the controller's own post-control friction velocity.

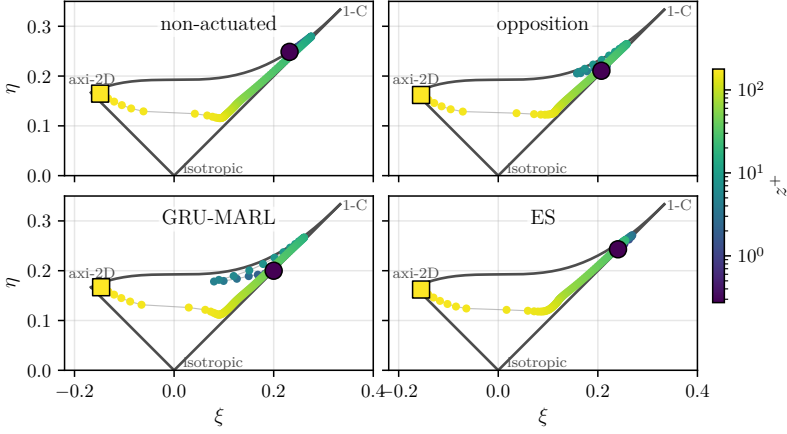


Figure 3: Anisotropy-invariant trajectories through the wall-normal coordinate.

velocity fluctuations, so its actuation follows the footprint of the sweeps and ejections at the detection plane by construction. The learnt actuation instead aligns with the streamwise low- and high-speed streaks. Figure 5 makes the relationship explicit: the mean actuation conditioned on the local streamwise and wall-normal fluctuations varies principally with u' and less strongly with w' . Across the learnt controllers the actuation correlates with u' at about 0.5 to 0.6, two to four times stronger in magnitude than its correlation with w' . The learnt actuation thus tracks the streamwise streaks, and opposes the wall-normal velocity. We describe this spatial relationship only, and do not address the order in which the actuation and the structures influence one another.

The joint probability density of (u', w') at the detection plane, in figure 7, shows how the two learnt controllers redistribute the momentum-carrying events. In the uncontrolled flow the ejections and sweeps contribute almost equally. Under every controller the sweep contribution exceeds the ejection contribution, so the ejections are weakened preferentially, and the effect is strongest for the evolution-strategy controller, whose ejection contribution to the inner-scaled Reynolds shear stress $\langle u'w' \rangle^+$ falls to about -0.10 , from -0.32 in the uncontrolled flow. These are contributions to the shear stress, not the quadrant occupation probabilities annotated in figure 7.

Figure 8 shows the premultiplied spectra of the streamwise and wall-normal velocity

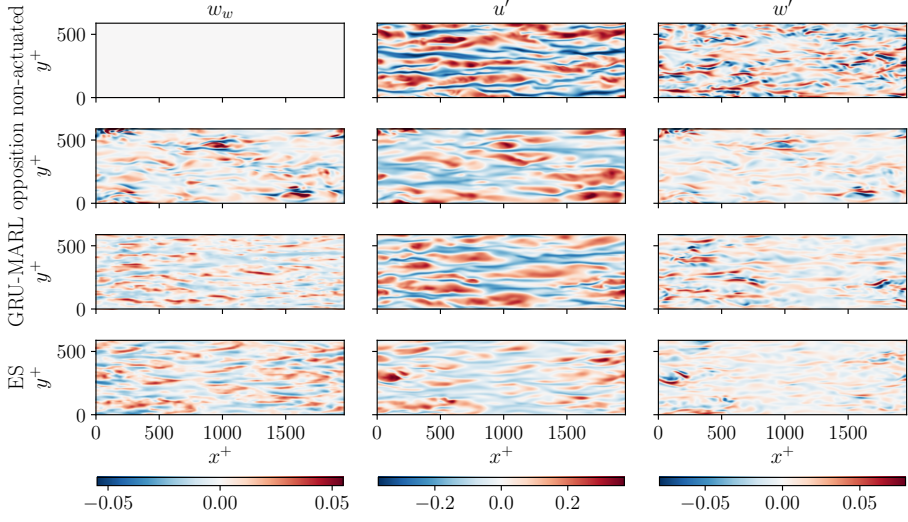


Figure 4: Instantaneous near-wall fields and wall actuation for the four cases.

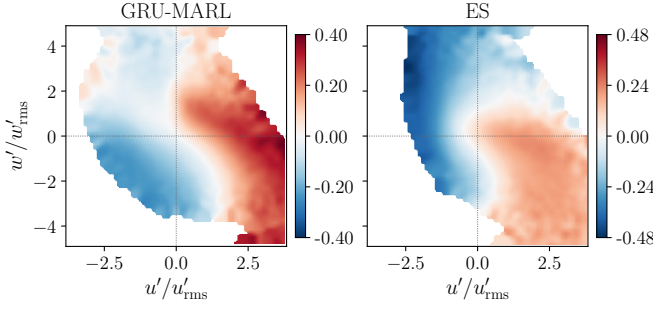


Figure 5: Conditional policy maps: mean actuation over (u', w') bins.

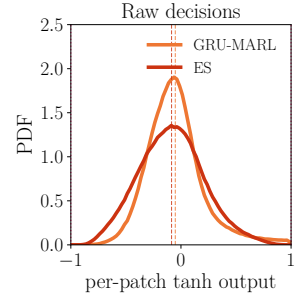


Figure 6: Actuation value distributions.

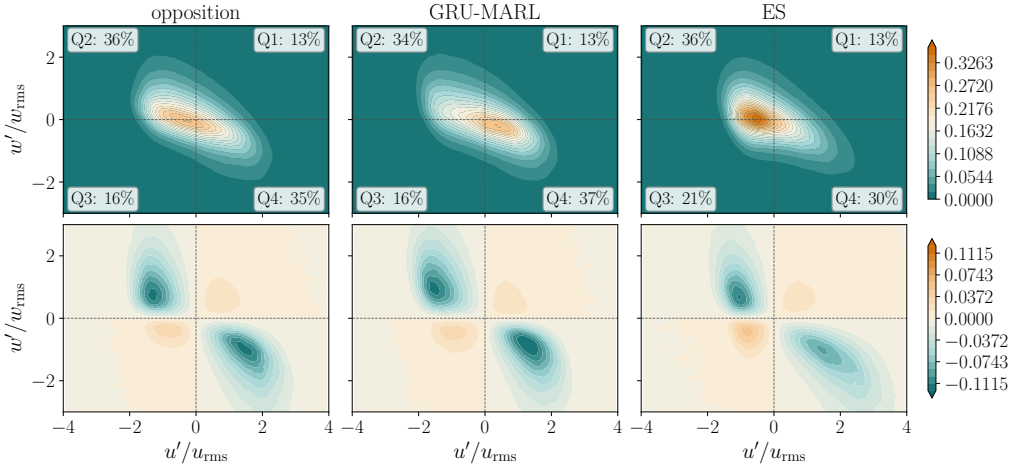


Figure 7: Joint PDF of (u', w') at the detection plane: raw (top) and premultiplied by $u'w'$ (bottom). Boxes give quadrant occupation fractions.

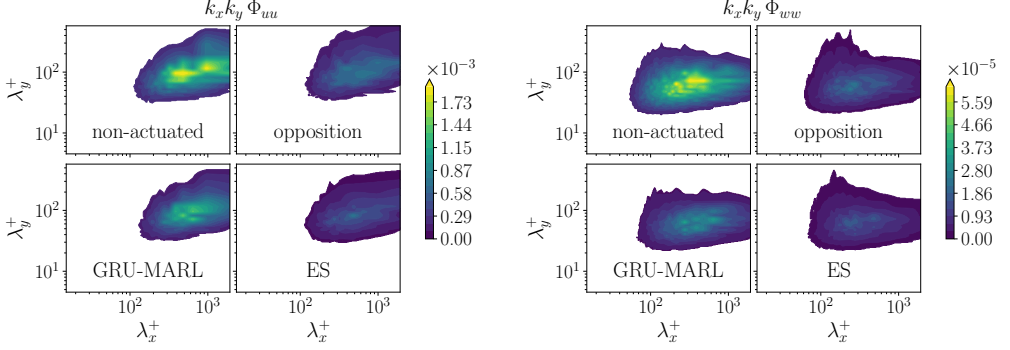


Figure 8: Premultiplied energy spectra near the wall, at $z^+ \approx 0.25$.

fluctuations near the wall. The energy of the controlled flows is modified at the spanwise wavelengths of the near-wall streaks, around $\lambda_y^+ \approx 100$ (deI Álamo *et al.* 2004), and the learnt controllers concentrate their effect there. The streamwise streak energy, peaked near $\lambda_x^+ \approx 700$ in the uncontrolled and GRU-MARL flows, is shifted to shorter wavelengths under opposition and most strongly redistributed towards small scales by the evolution strategy, whose long streaks are broken into shorter fragments (figure 4).

3.3. Smoothness and statistical steadiness of the learnt control

The comparison above is meaningful only if the learnt actuation controls the turbulence rather than settling into a fixed pattern set by the box. Figure 6 shows the distribution of wall-actuation values for each controller: it is smooth and stays away from the amplitude bound, with a root-mean-square amplitude of about 0.016 and a negligible saturated fraction. The streamwise-averaged actuation also remains statistically stationary and does not lock into a standing wave in the spanwise direction. The actuation is, then, an unsaturated response that tracks the evolving flow, not the saturated, frozen forcing that an objective scored on pumping power alone would favour.

4. Discussion

An evolution strategy optimises a closed-loop wall controller directly on the large, fully turbulent channel at $Re_\tau \approx 180$, and recovers the drag reduction that the same architecture loses when a gradient-based multi-agent policy, trained on a minimal box, is transferred to that channel. The gain has a simple reading: the search scores each complete episode by a single energy-aware number and never assigns a separate reward to each wall patch, so the zero-net-mass constraint that links the patches becomes part of how the actuation is formed rather than a source of interference in the optimisation, and the controller is tuned on the same flow it will drive. It reaches a drag reduction of about 26%, above the 17% retained by the transferred minimal-box policy and marginally above the 22% of opposition control. The added cost is a population of rollouts at each generation, which can run in parallel because they are independent.

The four flows differ substantially in structure, and the three controlled states remain distinct even where their drag reductions are similar. Opposition and the evolution strategy remove comparable amounts of skin friction, yet the diagnostics place their flows in different parts of the buffer layer, with different Reynolds-stress profiles, each scaled by its own friction velocity, and different anisotropy. The evolution-strategy actuation aligns predominantly with

the streamwise streaks, whereas opposition acts against the wall-normal velocity (Choi *et al.* 1994; Hammond *et al.* 1998; Stroh *et al.* 2015); in redistributing the momentum-carrying events at the detection plane it weakens the ejections preferentially and concentrates its effect near the spanwise streak wavelength $\lambda_y^+ \simeq 100$. Spanwise forcing produces a comparable lateral reorganisation of the streaks (Touber & Leschziner 2012), though through an open-loop rather than a sensed actuation.

The clearest separation is in the streaks themselves. The instantaneous fields (figure 4) and the streamwise spectra (figure 8) show opposition mainly lowering the streak amplitude, whereas the evolution strategy breaks the long streaks into shorter fragments and moves the streamwise energy to smaller scales. That two controllers reaching similar drag reduction act so differently on the coherence of the near-wall cycle suggests different mechanisms, distinguished by the flow topology rather than by the drag figure alone. Because the controller is smooth and was optimised on the same large domain on which it is assessed, this separation reflects the controlled turbulence and not the size of a training box.

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